Internet of Things & Edge Computing Architecture

Course: Internet of Things | Semester 8 | Contributor: Imlirenla Jamir

MQTT vs CoAP protocols, sensor interfacing, ESP32 microcontrollers, and edge AI inference.

IoT Communication Protocols

* MQTT publish/subscribe messaging: broker architecture, QoS levels 0, 1, 2, and keep-alive pings.
* CoAP Constrained Application Protocol: RESTful UDP-based communication for low-power nodes.
* LoRaWAN, Zigbee, and BLE long-range low-power sensor network standards.

Edge Computing & Hardware Interfaces

* GPIO, I2C, SPI, and UART serial communication bus protocols and clock timing.
* Edge computing vs centralized cloud: latency reduction, bandwidth conservation, and privacy.
* TinyML and quantized neural network deployment on microcontrollers.

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